

11 May 2026

Editorial Office
IEEE Access

Dear Editorial Office,

I am submitting the manuscript titled “Beyond English-Only GRPO: A Multi-Seed Empirical Study at Sub-3B Scale” for consideration in IEEE Access.

This work provides a multi-seed empirical study of Group Relative Policy Optimization (GRPO) post-training of a 1.5B distilled reasoning model under a single-GPU LoRA constraint. Building on the recent observation by Hochlehnert et al. (COLM 2025) that small-model RL reasoning gains may not survive multi-seed evaluation, we test three training-condition arms across three random seeds plus a single-seed mechanism ablation. The manuscript makes four contributions:

1. Multi-seed empirical evidence that vanilla English GRPO at sub-3B + LoRA scale exhibits $\sigma = 11.3$ percentage points (pp) seed variance on AMC-23 and consistently degrades AIME-2024 pass@1, confirming single-seed claims unreliable at this scale.
2. Documentation of the auxiliary-reward arm (A3) achieving the only positive AIME-2024 maj@8 effect ($+4.4 \pm 1.9$ pp over the untrained base), robust across three seeds.
3. A constant-bias ablation (A4) that partially refutes a pure reward-magnitude interpretation of the auxiliary-reward effect.
4. A 10-language MGSM evaluation showing null cross-lingual transfer for all training conditions (± 0.5 pp of base mean).

The work fits IEEE Access’s broad scope on engineering applications of machine learning, with particular relevance to deployment-constrained post-training of language models. Our reproducibility commitment (full code released under Apache-2.0, LoRA adapters, evaluation JSONs with SHA256 manifest, per-step training logs) addresses the IEEE emphasis on rigorous and verifiable research.

In compliance with IEEE Access policies on AI-assisted academic work, the manuscript and accompanying source code were prepared with assistance from Anthropic Claude (via the Claude Code command-line interface). All empirical results were independently produced and verified by the author, who bears sole responsibility for the technical claims, methodology, and any errors. The complete AI-assistance scope is detailed in the manuscript and in the VERIFICATION.md audit document of the released codebase.

The corresponding author and sole author is Vu Dang, an independent researcher based in Vietnam. The work has not been published elsewhere and is not under review at any other venue. Code and data are publicly available at <https://github.com/vudang4494/xling-grpo-sub3b> and permanently archived at Zenodo DOI 10.5281/zenodo.20061328.

Thank you for your consideration.

Sincerely,

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